

Final Project Description and Rubric

Semester	202510	Division	Computer Information Science
Assessment title in Syllabus	Final Project	Program	Bachelor of Data Science

Course Code	CIS 2013		
Course Title	Ethics and Sustainability for Computing		
CLOs	CLO1, CLO3, CLO4, CLO5	Accreditation Body	CAA & CIPS
Course Instructor	Alain Tcheukam	CRN	11899
Assessment Weight	40%	Submission Date	Week 15

For Group Work submissions an additional individual assessment will be conducted.

Grades for the students in one group will vary based on the individual performance in the additional assessment.

Student Declaration :

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Section No.	Deliverable 1 (Proposal – Ethical Consideration)	Deliverables 2, 3 & 4 (Ethical behaviors, moral issues, stakeholders, sustainability, processes)	Presentation + Oral Defence	Total	%
Marks Allocated	12.5	37.5	50	100	40



Marks Obtained					
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Chatbot (e.g. ChatGPT) statement

This submission must be completed without the use of any AI tools. Any violation of this requirement will be considered an attempt at plagiarism and will be addressed in accordance with the HCT Academic Integrity Policy.



An Ethics and Sustainability Policy for Noon E-Commerce: A Framework for Responsible Data Management and Sustainable Operations

CIS 2013 Final project

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PROJECT OBJECTIVES

This Report aims to achieve the following specific objectives:

1. **To Identify and Analyze Key Ethical Challenges:** To examine Noon's data management lifecycle and identify the core ethical issues related to data collection, storage, and usage, with a particular focus on surveillance, algorithmic fairness, and third-party accountability.
2. **To Define the Scope of a Comprehensive Policy:** To propose a preliminary scope for a new Ethics and Sustainability Policy that addresses three critical areas: data governance, operational sustainability, and supply chain accountability.
3. **To Ensure Legal and Ethical Alignment:** To establish how the proposed policy will align with key legal frameworks, primarily the UAE Personal Data Protection Law (PDPL), while also integrating global best practices inspired by regulations like the GDPR.
4. **To Map the Stakeholder Ecosystem:** To identify all key internal and external stakeholders, analyze their distinct ethical roles and responsibilities, and understand the implications of their actions on Noon's ethical performance.
5. **To Develop a Final, Actionable Policy:** To consolidate all analysis into a complete and detailed Ethics and Sustainability Policy for Noon, featuring specific commitments, clear accountability structures, and measurable monitoring mechanisms.

INTRODUCTION

The way businesses operate has been completely changed by the digital economy, and in the Middle East, Noon E-Commerce is a leading example of this major transformation. As one of the largest e-commerce platforms in the region, Noon processes millions of customer transactions every day. To make this possible, the company collects and uses huge amounts of personal data. At the same time, its operations, from the energy needed to run its data centers to the fuel used by its delivery fleet, create a significant environmental footprint.

For these reasons, developing a strong Ethics and Sustainability Policy is now a critical task for Noon. This is not just about being a good corporate citizen; it is a strategic necessity. In today's world, customers are more aware than ever about how their data is used and expect companies to be responsible. A strong policy is essential for Noon to maintain customer trust, to ensure it complies with important regulations like the UAE's data protection laws, and to secure its success for the future.

This report presents a comprehensive analysis and a detailed policy framework designed to guide Noon toward becoming a leader in ethical and sustainable business practices. It will systematically address the most important issues by examining the company's operations in three main areas. First, it will analyze the complex network of stakeholders including customers, employees, and partners, and define their ethical responsibilities. Second, it will evaluate the intersection of sustainability practices and legal requirements, with a special focus on Noon's use of adaptive AI systems and data protection laws. Finally, the report culminates in a proposed Ethics and Sustainability Policy that provides clear, actionable goals and a strong governance structure for Noon to follow.

COMPANY OVERVIEW

Noon.com is a major e-commerce company in the UAE, and it represents a big success story in the Middle East's digital growth. The company was started in 2016 by the well-known Emirati businessman Mohamed Alabbar, with a large investment from Saudi Arabia's Public Investment Fund (PIF) (Tracxn, 2025). From the beginning, Noon was created to be the leading Arabic-first online marketplace in the region. Today, it operates in the UAE, Saudi

Arabia, and Egypt and is considered by experts to be "the region's leading e-commerce platform" (Redseer, 2025).

What makes Noon special is that it is more than just an online store. It has grown into a complete digital ecosystem with many connected services. This includes noon.com for general shopping, "**Noon Food**" for restaurant delivery, "**noon Minutes**" for fast grocery delivery, and "**Noon Pay**" for digital payments. The company's main goal is "to become the engine of efficiency and convenience for the people we serve" (Noon, 2025). To achieve this goal, Noon must collect and use a very large amount of customer data.

At its heart, Noon is a data-driven company. Its success depends on collecting and analyzing different types of data to improve its services and give customers a personal experience.

According to its own privacy policy, the main types of data it collects are (Noon, 2025):

- **Customer Data:** This includes personal information like names and addresses, as well as behavioral data like what a customer searches for and buys.
- **Transactional and Financial Data:** This includes payment details for every purchase and the customer's order history.
- **Technical and Device Data:** The platform also collects technical information like IP addresses and the type of device being used, which helps with security and analysis.

For a large company like Noon, managing this data ethically is extremely important. It is not just about following the law; it is a key part of its business strategy. There are three main reasons why this is so important. First, it is essential for maintaining customer trust, which is the foundation of any successful e-commerce business. Second, it ensures regulatory compliance with important laws like the UAE's Personal Data Protection Law (PDPL), which helps the company avoid large fines (Centre Systems Group, 2024). Finally, in a very competitive market, showing a strong commitment to protecting data helps Noon stand out and attract customers who care about their privacy (HeyData, 2024).

CLO1: ETHICAL ISSUES IDENTIFICATIONS

As a leading e-commerce platform, Noon's business model is built on the collection, storage, and use of large amounts of customer data. While this data is essential for improving services, it also creates several important ethical challenges that need to be addressed. This section identifies the key ethical issues at each stage of Noon's data handling process.

1. Ethical Issues in Data Collection

The first ethical problem happens when data is collected. According to Noon's own privacy policy, the company automatically collects a lot of technical information, such as IP addresses, location, and device type, using tools like "cookies, tracking pixels, web beacons, etc." (Noon, 2025). This creates what some experts call a "surveillance infrastructure" that monitors how users interact with the platform. The main ethical issues here are **consent** and **transparency**. Customers usually give their consent by agreeing to long and complicated privacy policies that are difficult to understand. This means their consent may not be truly "informed" (Verpex, 2024). Another issue is **data minimization**, which is the principle of only collecting data that is absolutely necessary. Noon may be collecting more data than it needs for a simple purchase, which increases the privacy risk for customers.

2. Ethical Issues in Data Storage

Once the data is collected, the next ethical challenge is how to store it safely. The biggest concern here is **data security**. Noon has a major ethical and legal duty to protect its customers' data from being stolen or accessed by unauthorized people. Storing huge amounts of data makes the company a big target for hackers. A second important issue is **data retention**, which means how long the company keeps the data. Keeping customer data forever, even for future use, increases the risk of it being exposed in a data breach and goes against privacy principles. A good ethical policy should have clear rules for how long data is kept and ensure it is deleted securely when it is no longer needed.

3. Ethical Issues in Data Usage

The most complex ethical problems appear in how Noon uses the data it has collected. The company uses **profiling** to create detailed profiles of its customers to personalize their shopping experience. While this can be helpful, it also raises serious **privacy** concerns because it can feel like a form of surveillance. This data is then used by AI systems for **automated decision-making**, such as setting different prices for different people or deciding which products to show them. These AI systems can be like "black boxes," which means it is hard to know how they make decisions. This creates a risk of unfairness or discrimination (Blog.NewEcom.AI, 2024). Finally, **data sharing** with third-party companies, like logistics partners and marketing firms, creates a big **accountability** problem. Even if Noon is careful with data, it is hard to be sure that all its partners are just as careful. Noon is still ultimately responsible for protecting its customer data, even when it is in the hands of another company (Corridalegal, 2023).

CLO1: PRELIMINARY SCOPE OF ETHICAL POLICY

Based on the ethical challenges identified in Noon's operations, such as surveillance, unfair algorithms, and third-party risks, this report proposes a comprehensive **Ethics and Sustainability Policy**. This section outlines the initial scope of that policy. It defines the main goals, the guiding principles, and the key areas that the policy will focus on to make Noon a more responsible company.

1. Objectives and Guiding Principles

The primary goal of this policy is to establish Noon as the most trusted and responsible e-commerce platform in the Middle East. To achieve this, the policy will be guided by a clear set of objectives and principles.

Objectives: The policy has four main goals: The first is **To Protect Customers** by ensuring the highest level of privacy and security for their data. The second is **To Ensure Fairness** by making sure all customers are treated equally by our automated systems. The third is **To Be Accountable** by creating clear responsibility for ethical behavior within Noon and among its partners. Finally, the fourth objective is **To Be Sustainable** by reducing the company's negative impact on the environment.

Guiding Principles: These objectives will be achieved by following four core principles in everything we do:

- **1) Transparency**, meaning we will be open and honest about how we use data; **2) Accountability**, meaning we will take responsibility for our actions;
- **3) Data Minimization**, meaning we will only collect the data we absolutely need; and **4) Privacy by Design**, meaning we will build privacy protections into our systems from the very beginning.

2. Key Focus Areas for Policy Development

To turn these principles into real action, the detailed policy will concentrate on three main areas. These areas are designed to directly address the ethical issues we identified earlier.

1. **Data Governance and AI Ethics:** This area will create strict rules for how customer data is collected, used, and protected. It will also establish a framework to ensure that Noon's AI systems (used for pricing, recommendations, etc.) are fair, transparent, and free from bias.
2. **Third-Party and Supply Chain Accountability:** This area will focus on Noon's partners. A "Supplier Code of Conduct" will be created, which will require all logistics partners and marketplace sellers to follow the same high standards for data protection and ethical conduct that Noon follows.
3. **Operational and Environmental Sustainability:** This area will address Noon's physical impact on the environment. It will include policies for "Green IT" (such as using energy-efficient data centers), sustainable packaging, and low-carbon delivery methods (like using electric vehicles).

This structured scope provides a strong foundation for the detailed policy framework that will be presented later in this report.

CLO1: ALIGNMENT WITH LEGAL AND REGULATORY FRAMEWORK

A core part of the proposed ethics policy is its strong alignment with the laws and regulations in the UAE. Following the law is the minimum requirement for any ethical company, and this policy is designed to ensure Noon's operations are both responsible and legally sound.

The most important law for Noon to follow is the **UAE Personal Data Protection Law (PDPL)**. This policy requires strict obedience to the PDPL's main rules. This includes having clear ways to get customer **consent**, protecting customers' **rights** (the right to see or delete

their data), and having a strong plan to notify the UAE Data Office and customers in case of a **data breach (Centre Systems Group, 2024)**. Additionally, for financial services like "Noon Pay," the policy must follow the strict guidelines from the **Central Bank of the UAE** to keep all payment information safe.

However, being a true ethical leader means doing more than just what the law requires. For this reason, the policy also includes global best practices and ethical standards. It takes inspiration from the principles of the European Union's **General Data Protection Regulation (GDPR)**, which is seen by many as the best data protection standard in the world (**PrivacyEngine, 2023**). This includes using principles like **"Privacy by Design"** (building privacy into systems from the start) and **"Data Minimization"** (collecting only the data that is absolutely needed).

This two-part approach strictly following the law while also aiming for higher ethical standards ensures that Noon not only avoids legal problems but also builds a culture of trust and responsibility, strengthening its reputation as a leader in the region's digital economy.

CLO3: STAKEHOLDERS AND ETHICAL RESPONSIBILITIES

To create an effective ethics policy, it is crucial to understand all the different groups of people, or **stakeholders**, who are connected to Noon's operations. These stakeholders are not just watching from the side; their actions and expectations directly shape how ethical the company is. We can divide them into two main categories.

First, there are **internal stakeholders**. This group includes the employees who build and run the platform, such as **developers** and **data scientists**, as well as **management** and the company's **owners**. Second, there are **external stakeholders**. This is a larger group that includes **customers**, **vendors** (the sellers on Noon's marketplace), **suppliers** (like logistics and packaging companies), and **regulators** (such as the UAE Data Office).

Each of these groups has a unique and important ethical role. For example, developers and data scientists have an ethical duty to build fair and secure systems, following the principle of **"Privacy by Design."** Management is responsible for creating and enforcing the rules that ensure everyone acts ethically. Externally, customers have a right to **privacy** and

transparency, while vendors and suppliers have a responsibility to handle any shared customer data securely and to follow fair labor practices.

The actions of each stakeholder have a direct impact on Noon's ethical performance, and a failure by any one group can cause a major problem. For instance, if a developer creates a pricing algorithm that is accidentally biased, it can lead to unfair treatment of certain customers. This is an ethical failure. Similarly, if a third-party logistics supplier has weak data security and loses customer data, it directly harms Noon's reputation and violates customer trust, even if Noon's own systems are secure. This shows that ethical responsibility is shared across the entire network. Therefore, the proposed policy must clearly define these roles and create a system of shared accountability to ensure that every stakeholder, from the company's owners to its delivery partners, is committed to the same high ethical standards.

CLO4: ADAPTIVE AND AI SYSTEMS

To stay competitive and efficient, Noon's business relies heavily on **Adaptive AI Systems**. These are advanced computer programs that can learn from new data and change their own actions without a human programmer. The main use of these AI systems at Noon is to create a personalized shopping experience for every customer and to make its logistics operations run smoothly. For example, AI is used in **recommendation engines** that suggest products you might like, in **dynamic pricing** where prices can change based on demand, and in **optimizing delivery routes** to make them faster. These systems are called "adaptive" because they are always learning and improving on their own.

However, this powerful technology creates serious ethical challenges, especially around **accountability** and **fairness**. A major issue is the lack of transparency, often called the **"black box" problem**. This means it can be very difficult to understand why the AI made a certain decision. This lack of clarity can lead to unfair outcomes, such as accidental price discrimination against certain groups of customers or creating "filter bubbles" that limit the products a user is shown (**Blog.NewEcom.AI, 2024**).

This raises the most important ethical question: who is responsible when an AI system makes a mistake or causes harm? Is it the developer who wrote the code, the company that uses the system, or the AI itself? To address this, Noon's policy must focus on governing its AI, not

just using it. This means establishing a clear framework for **human oversight** of important decisions, requiring the use of **Explainable AI (XAI)** for high-stakes situations (like credit scoring for Noon Pay), and creating an **AI Ethics Committee** to regularly check these systems for bias. Without strong governance, the very adaptability that makes these AI systems so powerful also makes them a major source of ethical and reputational risk for the company.

CLO4: COMPUTER FORENSICS

In an environment where data breaches and digital fraud are persistent threats, the principles of **computer forensics** are essential for establishing accountability and ensuring a secure platform. The primary **use of computer forensics** for Noon is not just for post-incident investigation but for maintaining a state of "forensic readiness." This means having the policies, tools, and trained personnel in place to respond to a security breach swiftly and effectively. In the event of an incident, such as unauthorized access to customer data, the forensics team would be responsible for collecting, preserving, and analyzing digital evidence while maintaining a strict chain of custody to ensure the findings are admissible in legal proceedings. This entire process relies on the effective **use of digital trails**, which are the electronic footprints left by every action on the platform. These trails include everything from server access logs and IP addresses to transaction records and file modification histories. For Noon, these digital trails serve a dual purpose: defensively, they are crucial for tracing the steps of an attacker and identifying system vulnerabilities; and proactively, they are used for fraud detection and monitoring for internal misconduct. However, the existence of such extensive digital trails creates a significant ethical responsibility. The policy must therefore mandate that these trails are securely stored, access is strictly limited to authorized personnel for legitimate security purposes, and they are disposed of according to a defined data retention schedule to protect customer privacy from potential misuse.

CLO5: SUSTAINABILITY POLICY

To address its environmental and social impact, Noon requires a formal and actionable **sustainability policy**. This policy is designed to be comprehensive, integrating sustainability into the core of its operations. It is structured around three distinct yet interconnected pillars: social responsibility, IT governance, and procurement.

First, a Socially Responsible Policy for the Firm will be implemented. This policy establishes Noon's commitment to the well-being of its employees, partners, and the communities it serves. It moves beyond basic labor compliance to create a positive social impact. Key commitments include enforcing a strict **Supplier Code of Conduct** for all logistics and warehouse partners, which will mandate fair wages and safe working conditions. The company will also launch a **"Local First" initiative** to promote local small and medium-sized businesses on its platform.

Second, a Sustainability Policy for IT Governance, often called "Green IT," will directly address the environmental footprint of Noon's digital infrastructure. As a technology company, minimizing the impact of its data centers is a critical responsibility. Key commitments include prioritizing cloud providers that use **100% renewable energy** (Kearney, 2024) and establishing a formal **e-waste management program** to responsibly dispose of all retired IT hardware, in line with national guidelines (UAE Ministry of Finance, 2025).

Third, a Sustainable Procurement Policy for the Firm will govern the physical materials used in Noon's daily operations, from packaging to delivery. This policy aims to create a circular and low-carbon supply chain. Key commitments include using **100% sustainable packaging** by a set deadline and beginning a transition to **electric vehicles (EVs)** for its delivery fleet. This is a key strategy for reducing the carbon footprint in e-commerce (DHL, 2024). Additionally, all major suppliers will be required to meet minimum **Environmental, Social, and Governance (ESG) criteria**, making sustainability a key factor in all procurement decisions.

CONCLUSION

This report has provided a comprehensive analysis of the ethical and sustainable responsibilities facing Noon E-Commerce, creating a detailed policy framework to guide the company toward responsible leadership in the digital age. The summary of the main ethical insights shows that Noon's core challenges—such as surveillance, algorithmic bias, and third-party accountability—are not just technical problems. Instead, they are deeply connected to the relationships between all stakeholders, from developers to customers and suppliers. The analysis confirms that to maintain customer trust and succeed in the long term, Noon must do more than just follow the law; it must build ethical principles like fairness, transparency, and accountability into every part of its business.

To build on this foundation, this report offers two key future recommendations that would position Noon as a true innovator in corporate responsibility in the region:

First, Launch a Public AI Explainability Portal. Noon should create a user-friendly portal where customers can get simple, real-time explanations for automated decisions, such as why they received a certain product recommendation or price. This would set a new standard for transparency and give customers meaningful control and insight into how their data is used.

Second, Commit to Full Fleet Electrification. Noon should announce a bold, time-bound plan to transition 100% of its last-mile delivery fleet to electric vehicles. This decisive action would not only drastically reduce its carbon footprint but also establish Noon as a leader in sustainable logistics in the Middle East.

By adopting the comprehensive policies outlined in this report and pursuing these forward-thinking recommendations, Noon can effectively mitigate its risks, strengthen its brand reputation, and build a truly sustainable business that creates profound and lasting trust with its customers, investors, and the wider community.

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